12TH BIOLOGY · CHAPTER 19 · SOLVED EXERCISE

# Chromosome and DNA

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

## SECTION 1: MULTIPLE CHOICE QUESTIONS

**16. Which of the following looks like beads on a string?**

(a) Chromosomes (b) Chromatin
(c) Nucleosomes (d) Heterochromatin

**Answer: (c) Nucleosomes** — Nucleosomes joined by linker DNA resemble beads on a string (nucleosome string, 11 nm thick).

**17. The most convincing proof that DNA is the genetic material was provided by;**

(a) Mendel (b) Morgan
(c) Hershey & Chase (d) Meselson and Stahl

**Answer: (c) Hershey & Chase** — They showed that the DNA of a bacteriophage, not its protein, enters the bacterium and directs the making of new phages.

**18. Degeneracy of genetic code means that;**

(a) A codon consists of 3 nucleotides
(b) Multiple codons can specify the same amino acid
(c) Same codon can specify multiple amino acids
(d) Each codon is read independently of the adjacent codons

**Answer: (b) Multiple codons can specify the same amino acid** — This is the textbook definition of degeneracy; (a) is the triplet nature and (d) is the non-overlapping nature.

**19. How many codons are needed to specify for five amino acids?**

(a) 3 (b) 5
(c) 15 (d) 30

**Answer: (b) 5** — One codon specifies one amino acid, so five codons (15 nucleotides) are needed.

**20. Which of the following is NOT involved in the initiation of replication?**

(a) DNA ligase (b) DNA gyrase
(c) single-stranded binding protein (d) Primase

**Answer: (a) DNA ligase** — DNA ligase works at termination, when it joins the Okazaki fragments.

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**21. Why DNA polymerase synthesises the leading and lagging strands of DNA in different manners?**

(a) The origins of replication occur only at the 5’ end
(b) DNA ligase works only in the 3’ to 5’ direction
(c) DNA polymerase can only work on one strand at a time
(d) DNA polymerase can join new nucleotides only to the 3’ end of strand

**Answer: (d) DNA polymerase can join new nucleotides only to the 3’ end of strand** — The two template strands run in opposite directions, but the enzyme can build only in the 5′ to 3′ direction.

**22. Synthesis of a new DNA strand usually begins with:**

(a) RNA primer (b) DNA primer
(c) DNA ligase (d) Okazaki fragment

**Answer: (a) RNA primer** — Primase first attaches a short primer of 5–10 RNA nucleotides on the template.

**23. The elongation of the leading strand during DNA synthesis;**

(a) Progresses away from the replication fork
(b) Occurs in the 3’ – 5’ direction
(c) Produces Okazaki fragments
(d) Depends on the action of polymerase

**Answer: (d) Depends on the action of polymerase** — DNA polymerase III builds it continuously toward the fork in the 5′ to 3′ direction; (a) and (c) describe the lagging strand.

**24. Which of the following enzymes remove supercoiling in replicating DNA ahead of the replication fork?**

(a) DNA polymerases (b) Helicases
(c) Primases (d) Topoisomerases

**Answer: (d) Topoisomerases** — Gyrase (a topoisomerase) attaches ahead of the replication fork and prevents supercoiling.

**25. Which enzyme joins the bits of DNA?**

(a) DNA polymerase (b) DNA ligase
(c) Endonuclease (d) Primase

**Answer: (b) DNA ligase** — DNA ligase joins the Okazaki fragments of the lagging strand.

**26. Which of these subunits of RNA polymerase is required to initiate transcription?**

(a) Alpha (α) (b) Sigma (σ)
(c) Omega (ω) (d) Beta (β)

**Answer: (b) Sigma (σ)** — The σ subunit recognizes the promoter and initiates synthesis.

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**27. When and where does the splicing of mRNA occur?**

(a) Before transcription in nucleus
(b) After transcription in nucleus
(c) After transcription in cytoplasm
(d) After mRNA attachment with ribosome

**Answer: (b) After transcription in nucleus** — It is a post-transcriptional modification, done before the mRNA leaves the nucleus.

**28. What happens in mRNA splicing?**

(a) Introns and exons are removed from the gene
(b) Introns and exons are joined together
(c) Exons are removed and introns are joined together
(d) Introns are removed and exons are joined together

**Answer: (d) Introns are removed and exons are joined together** — Introns do not code for protein, so they are removed and the remaining exons are joined.

**29. Which of these happens first during protein synthesis?**

(a) Peptide bond formation
(b) Binding of aminoacyl t-RNA with ribosome
(c) Translocation
(d) Binding of release factor with ribosome

**Answer: (b) Binding of aminoacyl t-RNA with ribosome** — In initiation, the first aa-tRNA binds to the smaller ribosomal subunit; the other three events come later.

**30. The synthesis of protein chains is initiated with the amino acid;**

(a) Arginine (b) Methionine
(c) Serine (d) Valine

**Answer: (b) Methionine** — The start codon AUG specifies methionine (N-formyl-methionine in prokaryotes).

**31. The main function of tRNA is to:**

(a) Bind the mRNA with the smaller ribosomal subunit
(b) Form a portion of ribosomes
(c) Bind the complementary RNA and DNA strands together
(d) bring amino acids from the cytoplasm to the ribosomes?

**Answer: (d) bring amino acids from the cytoplasm to the ribosomes?** — Each aa-tRNA delivers its specific amino acid to the ribosome.

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**32. Ultraviolet radiation can cause mutations by;**

(a) Making double covalent bonds between adjacent pyrimidine bases
(b) Breaking phosphodiester bonds in DNA strands
(c) Adding nucleotides in NA
(d) Substituting nucleotides in DNA

**Answer: (a) Making double covalent bonds between adjacent pyrimidine bases** — UV makes a double covalent bond between two adjacent pyrimidines (C and T); (b) is the effect of ionizing radiation.

**33. Phenylketonuria is a genetic disorder with a defect in the metabolism of:**

(a) Vitamin (b) Glucose
(c) Amino acid (d) Lipid

**Answer: (c) Amino acid** — The amino acid phenylalanine is not converted into tyrosine because the PAH enzyme is defective.

**34. The condition sickle cell anaemia is due to;**

(a) Deletion type point mutation (b) Substitution type point mutation
(c) Insertion type point mutation (d) Chromosomal rearrangement

**Answer: (b) Substitution type point mutation** — The DNA nucleotide T is substituted by A, so glutamic acid is replaced by valine.

*The MCQs are numbered 16–34 as printed in the textbook. In MCQ 32, option (c) is printed as "NA"; it should read "DNA".*

## SECTION 2: SHORT QUESTIONS

**17. What are the four types of chromosomes on the basis of position of centromere?**

| Type of chromosome | Position of centromere |
|---|---|
| Metacentric | At the centre |
| Sub-metacentric | Slightly away from the centre |
| Acrocentric | Near the end |
| Telocentric | At the end |

**18. What is chromatin fibre? How is it formed during condensation of chromatin?**

Chromatin fibre is a 30 nm thick fibre made of tightly packed nucleosomes. In the early stages of cell division, the nucleosome string (11 nm) coils again about its axis. The nucleosomes become tightly packed (condensed) and form the thicker chromatin fibre.

### 19. Briefly describe the concept of gene and gene locus.

A gene is a short segment of DNA which encodes the sequence of amino acids of a particular polypeptide. The location where a gene is present on a chromosome is called its locus (plural: loci). On homologous chromosomes, the alleles of a gene are present at the same locus.

### 20. State the functions of DNA polymerase I, II and III.

- **DNA polymerase I:** supports polymerase III; it removes the RNA primers at the end of replication and fills the gaps with DNA nucleotides.
- **DNA polymerase II:** repairs DNA damage during the lifetime of a cell.
- **DNA polymerase III:** the main enzyme of replication; it adds new nucleotides along both template strands in the 5′ to 3′ direction and proofreads them.

### 21. State the central dogma of gene expression.

The English biologist Francis Crick used the term “Central Dogma” for the basic mechanism of gene expression. It states that information passes from the genes (DNA) to an RNA copy of the gene, and the RNA copy directs the sequential assembly of a chain of amino acids:

DNA → RNA → Protein

The flow of genetic information is one-way. It consists of two steps: **transcription** (synthesis of RNA from DNA) and **translation** (synthesis of a protein by using the information in RNA).

### 22. What do you mean by “degeneracy of genetic code”?

Degeneracy of the genetic code means that multiple codons can specify the same amino acid. For example, GGC and GGU both specify glycine. This characteristic is important for mutation tolerance and evolutionary flexibility.

*The textbook’s example on p.132 says that AUG, GUG, UUG and CUG all code for methionine. In the standard genetic code only AUG codes for methionine, so the glycine example from p.144 of the same chapter is used here.*

### 23. Enlist some commonly occurring minor mutations in human.

- **Single nucleotide polymorphisms (SNPs):** involve a single nucleotide; very common; harmless or with subtle effects, e.g., inability to digest milk (lactose).
- **Insertions and deletions:** add or remove one or more nucleotides, e.g., cystic fibrosis is commonly caused by a deletion of just three nucleotides.
- **Silent mutations:** do not change the amino acid sequence, e.g., GGC to GGU, both code for glycine.
- **Missense mutations:** change only one amino acid, e.g., glutamic acid replaced by valine in haemoglobin (sickle cell anaemia).
- **Nonsense mutations:** create a premature stop codon, giving a short, non-functional protein, e.g., Duchenne muscular dystrophy.

## 24. Differentiate between:

### • Histone and nucleosome

| Histone | Nucleosome |
|---|---|
| A positively charged protein of the chromosome, rich in arginine and lysine. | An aggregation of DNA and histones. |
| Five types: H2A, H2B, H3, H4 and H1. | About 146 nucleotides of DNA wrapped twice around a histone octamer. |

### • Heterochromatin and euchromatin

| Heterochromatin | Euchromatin |
|---|---|
| The inactive portion of chromatin. | The active portion of chromatin. |
| It produces dark bands. | It does not produce dark bands. |

### • Conservative and semi-conservative model of DNA replication

| Conservative model | Semi-conservative model |
|---|---|
| The parent double helix remains intact. | The two strands of the parent DNA separate; each acts as a template. |
| One daughter DNA has both parent strands and the other has both new strands. | Both daughter DNAs have one parent strand and one new strand. |
| It was not supported by the Meselson–Stahl experiment. | It was proved by the Meselson–Stahl experiment. |

### • DNA helicase and DNA gyrase

| DNA helicase | DNA gyrase |
|---|---|
| An unwinding enzyme that breaks the hydrogen bonds between the base pairs of the two strands. | An enzyme (topoisomerase) that prevents supercoiling of DNA. |
| It works at the origin site and makes the replication forks. | It attaches ahead of the replication fork. |

### • DNA polymerase I and II

| DNA polymerase I | DNA polymerase II |
|---|---|
| It removes the RNA primers and fills the gaps with DNA nucleotides. | It repairs DNA damage. |
| It works at the end of replication. | It works during the lifetime of a cell. |

## • Leading strand and lagging strand of DNA

| Leading strand | Lagging strand |
|---|---|
| It grows toward the replication fork. | It grows away from the replication fork. |
| It is made continuously as one chain. | It is made discontinuously as Okazaki fragments. |
| It needs only one primer. | It needs a new primer for each Okazaki fragment. |

## • Translation and transcription

| Translation | Transcription |
|---|---|
| Synthesis of a protein by using the information present in mRNA. | Synthesis of a complementary RNA from DNA. |
| It is the second step of gene expression. | It is the first step of gene expression. |
| It occurs on ribosomes. | It occurs on the template strand of DNA, by RNA polymerase. |

## • Genetic code and codon

| Genetic code | Codon |
|---|---|
| The information in DNA for the synthesis of proteins. | A sequence of three nucleotides which specifies one amino acid. |
| It consists of a series of codons. | It is one unit of the genetic code; there are 64 codons. |

## • Codon and anticodon

| Codon | Anticodon |
|---|---|
| A sequence of three nucleotides on mRNA. | A sequence of three nucleotides on tRNA. |
| It specifies an amino acid. | It is complementary to the codon and matches it at the ribosome. |

## • Gene and allele

| Gene | Allele |
|---|---|
| A short segment of DNA which encodes the sequence of amino acids of a particular polypeptide. | An alternative form of a gene. |
| It is present at a particular locus of a chromosome. | Alleles are present at the same locus on homologous chromosomes. |

## • Intron and exon

| Intron | Exon |
|---|---|
| A region of DNA or mRNA that does not code for protein. | A region of DNA or mRNA that is expressed (translated into protein). |
| It is removed during splicing. | Exons are joined after splicing and form the mature mRNA. |

## • Start codon and stop codon

| Start codon | Stop codon |
|---|---|
| AUG is the start codon. | UGA, UAG and UAA are the stop codons. |
| It specifies methionine and begins protein synthesis. | It specifies no amino acid and stops protein synthesis. |

## • Non-sense codon and sense codon

| Non-sense codon | Sense codon |
|---|---|
| A codon that does not specify any amino acid. | A codon that specifies an amino acid. |
| There are 3: UGA, UAG and UAA. | There are 61, e.g., UUU for phenylalanine. |

## • Repressor and activator proteins

| Repressor proteins | Activator proteins |
|---|---|
| They bind to operators and block RNA polymerase. | They bind to operators and help RNA polymerase to bind to the promoter. |
| They stop or reduce transcription. | They start transcription or increase its speed. |
| This is negative gene regulation. | This is positive gene regulation. |

## • Point mutation and chromosomal mutation

| Point mutation | Chromosomal mutation |
|---|---|
| A change in only one or a few nucleotides. | A large-scale change in the position of genes, or in the structure or number of chromosomes. |
| Types: substitution, insertion, deletion. | Types: chromosomal rearrangements, insertional inactivation, non-disjunction. |
| Example: sickle cell anaemia. | Example: Down syndrome. |

### • Down syndrome and Klinefelter syndrome

| Down syndrome | Klinefelter syndrome |
|---|---|
| Non-disjunction in autosome pair 21. | Non-disjunction in the sex chromosomes. |
| 47 chromosomes with trisomy 21. | 47 chromosomes with XXY (44+XXY). |
| The child may be male or female. | The affected person is a sterile male. |

### • Klinefelter syndrome and Tuners syndrome

| Klinefelter syndrome | Turner syndrome |
|---|---|
| Sex chromosomes are XXY (trisomy). | Sex chromosomes are XO (monosomy). |
| 47 chromosomes (2n+1). | 45 chromosomes (2n–1). |
| Sterile male with some female characteristics. | Sterile female with short stature and webbed neck. |

*"Sense codon" is not defined in the textbook; the standard meaning is given. The last pair is printed as "Tuners syndrome"; it means Turner syndrome.*

## SECTION 3: LONG QUESTIONS

### 15. Annotate the detailed structure of a chromosome.

Chromosomes are thin, thread-like structures made of DNA and histone proteins. The number and shape of chromosomes are different in each species; this complete set is called the **karyotype**. In the body cells of diploid organisms, chromosomes occur in pairs called **homologous chromosomes**. When a cell is not dividing (interphase), the chromosomes remain uncoiled and form a fine network called **chromatin**. When the cell prepares to divide, the DNA coils around histones and condenses into thick, short chromosomes.

#### Morphology

- **Chromatids:** After duplication in interphase, each chromosome consists of two identical strands called sister chromatids.
- **Centromere:** The constricted region where the sister chromatids are attached. It is a specific DNA sequence of about 220 nucleotides.
- **Kinetochore:** A disc-shaped protein structure attached to the centromere. Here the chromosome attaches to the mitotic spindle during cell division.
- **Secondary constriction** (Nuclear Organizer Region, NOR): present in some chromosomes; it makes nucleoli during interphase.
- **Satellite:** The knob-like end of the chromosome beside the secondary constriction. It has highly repetitive DNA sequences called junk DNA.
- **Telomeres:** The terminal ends of chromosomes. They prevent two chromosomes from attaching with each other at their ends.

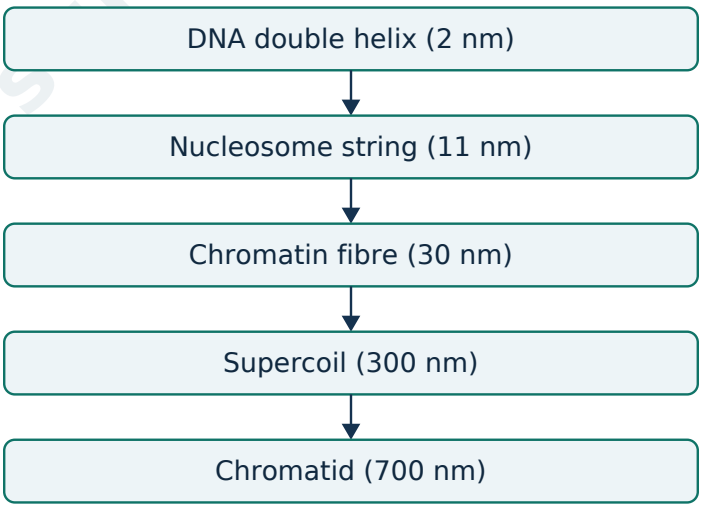
## Types on the basis of centromere position

| Type of chromosome | Position of centromere |
|---|---|
| Metacentric | At the centre |
| Sub-metacentric | Slightly away from the centre |
| Acrocentric | Near the end |
| Telocentric | At the end |

## Chemical composition and organization

A chromosome is made up of 40% DNA and 60% proteins (histones). The DNA of an average human chromosome is about 5 cm long and has about 140 million nucleotides. DNA is negatively charged due to its phosphate groups. It has a strong affinity for histones, which are positively charged due to basic amino acids such as arginine and lysine. The levels of organization are:

1. **Nucleosome string (11 nm):** About every 146 nucleotides of DNA (2 nm thick) wrap twice around a **histone octamer** (two copies each of H2A, H2B, H3 and H4). This forms a **nucleosome**. Nucleosomes are connected by linker DNA of about 80 nucleotides. The fifth histone, H1 (linker histone), lies at the base of the nucleosome and helps in winding the chain. The nucleosomes look like beads on a string.
2. **Chromatin fibre (30 nm):** In the early stages of cell division, the nucleosome string coils about its axis. The tightly packed nucleosomes make a thicker fibre called the chromatin fibre.
3. **Chromatid (700 nm):** The chromatin fibre folds upon itself to make a supercoil (300 nm thick). The supercoil coils and condenses further and takes the shape of a chromatid.




*Levels of organization of a chromosome*

### 16. Explain the concept of alleles as the alternative forms of a gene.

In 1909, the Danish botanist Wilhelm Johannsen introduced the term “gene” for the basic unit of heredity. A **gene** is a short segment of DNA which encodes the sequence of amino acids of a particular polypeptide. The whole DNA molecule of a chromosome does not carry messages for proteins; only short segments (genes) do.

The location where a gene is present on a chromosome is called its **locus** (plural: loci). The alternative forms of a gene are called **alleles**. On homologous chromosomes, alleles are present at the same locus. For example, if a chromosome has an allele for eye colour at a particular locus, its homologous chromosome will carry the allele of the same character at the same locus.

### 17. Narrate the experimental work of Griffith that proved that DNA is the hereditary material.

In 1928, the British biologist Fredrick Griffith worked on the pathogenic bacterium *Streptococcus pneumoniae*. It has two strains: the **S-type** has a polysaccharide coat and forms smooth colonies; the **R-type** lacks the coat and forms rough colonies.

| Injection given to mice | Result |
|---|---|
| Live S-type bacteria | Mice developed pneumonia and died (S-type is virulent) |
| Live R-type bacteria | Mice showed no effects |
| Heat-killed S-type bacteria | Mice remained healthy |
| Mixture of heat-killed S-type and live R-type | Mice developed pneumonia and died; live S-type bacteria were found in their blood |

**Conclusion:** The information for making the polysaccharide coat and for virulence had passed from the heat-killed S-type bacteria to the live R-type bacteria. So the R-type made a coat and changed into virulent S-type. Griffith called this process **transformation**, i.e., the transfer of genetic material from one organism to another to change the genetic makeup of the recipient. He could not find the chemical nature (DNA or protein) of the transferred material.

### 18. Narrate the experimental work of Hershey and Chase that proved that DNA is the hereditary material.

In 1952, two American biologists, Alfred Hershey and Martha Chase, experimented with **bacteriophages** (viruses that attack bacteria) to confirm that DNA is the genetic material. A bacteriophage has a DNA core and a protein coat. When it infects a bacterium, it injects some hereditary material which directs the production of new phages. The question was whether this material is DNA or protein.

1. Some phages were grown on a medium with radioactive sulphur ($^{35}S$). It was incorporated into the amino acids of their **protein coats**.
2. Other phages were grown on a medium with radioactive phosphorus ($^{32}P$). It was incorporated into the phosphate groups of their **DNA**.

3. **Setup 1:** phages with $^{35}$S-labelled protein were allowed to infect bacteria. **Setup 2:** phages with $^{32}$P-labelled DNA were allowed to infect bacteria.
4. After some time, the infected bacterial cells and the medium were separated by centrifugation and tested for radioactivity.

| Setup | Radioactivity found in |
|---|---|
| 1 ($^{35}$S-labelled protein) | The medium, not in the bacterial cells |
| 2 ($^{32}$P-labelled DNA) | The bacterial cells, not in the medium |

**Conclusion:** During infection, the DNA of the phage entered the bacterial cells and directed the making of new phages, while the protein coat remained outside. So DNA, not protein, is the hereditary material.

## 19. Describe the three models proposed about the mechanism of DNA replication.

1. **Conservative model:** The double helix of the parent DNA remains intact and generates a completely new double helix. So one daughter DNA contains both strands of the parent DNA and the other contains both new strands.
2. **Semi-conservative model:** The two strands of the parent DNA separate. Each strand acts as a template for the synthesis of a new strand. So both daughter DNAs consist of one parent strand and one new strand.
3. **Dispersive model:** The parent double helix is broken and dispersed into nucleotides, which combine with new nucleotides. So both daughter DNAs contain strands which are mixtures of old and new nucleotides.

## 20. Narrate the work of Meselson and Stahl to justify the semi-conservative replication.

In 1958, two American biologists, Mathew Meselson and Franklin Stahl, tested the three models and proved that DNA replication is semi-conservative.

1. They grew bacteria for many generations in a medium containing the heavy isotope of nitrogen ($^{15}$N). The DNA of these bacteria became denser than normal.
2. They transferred these bacteria to a medium containing normal $^{14}$N, so that any new DNA would be made with $^{14}$N.
3. They collected DNA at different intervals, dissolved it in cesium chloride solution and carried out ultracentrifugation. Each DNA settled where its density matched the density of the cesium; denser DNA moved further down.

| Sample | Sediment(s) observed | Explanation |
|---|---|---|
| 0 minutes (before replication) | One band of heavy density | Both strands of all DNA contain $^{15}$N |
| 20 minutes (after first replication) | One band of intermediate density | Each daughter DNA is a hybrid: one $^{15}$N strand and one $^{14}$N strand |

| Sample | Sediment(s) observed | Explanation |
|---|---|---|
| 40 minutes (after second replication) | Two bands: one intermediate and one light | Each hybrid DNA gave one hybrid DNA and one light DNA (both strands $^{14}N$) |

**Conclusion:** DNA replicates in a semi-conservative manner.

## 21. Describe the events of the process of DNA replication.

The basic mechanism of DNA replication is similar in all organisms. The following steps refer to prokaryotes.

### 1. Opening up the DNA double helix

- Replication begins at a specific sequence of nucleotides called the **replication origin site** (one in prokaryotes, many in eukaryotes). An initiator protein binds to it.
- **Helicases** break the hydrogen bonds between the base pairs. The strands separate and make two Y-shaped **replication forks** (together called a replication bubble).
- **Gyrase** (topoisomerase) attaches ahead of the replication fork and prevents supercoiling.
- **Single-strand binding proteins** bind to the single strands and prevent them from re-joining. Both strands act as templates.

### 2. Building a primer

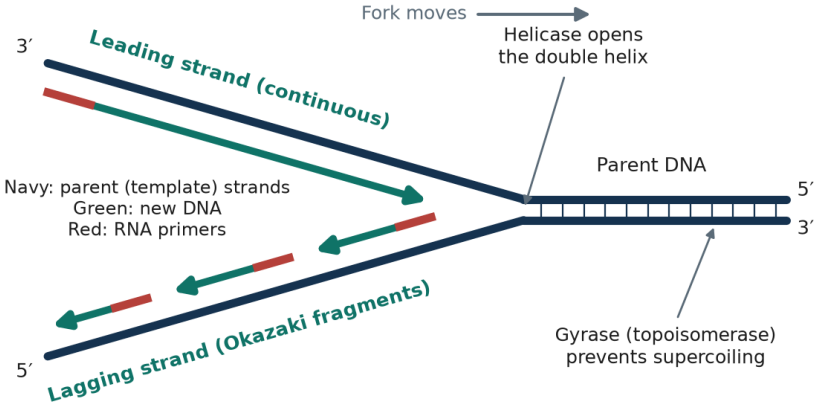
New DNA cannot be started on a bare template. So the enzyme **primase** attaches a short fragment of 5–10 RNA nucleotides, called the **primer**, on each template strand.

### 3. Assembling complementary strands (extension)

**DNA polymerase III** is the main enzyme. It matches free nucleotides to the template (A with T, C with G). The raw materials are deoxyribonucleoside triphosphates (dATP, dCTP, dGTP, dTTP); two phosphates break off and release the energy for building. The enzyme can add nucleotides only to the 3′ end, so the new strand grows in the 5′ to 3′ direction.

- **Leading strand:** On the template that runs toward the fork, the enzyme moves in the same direction as the unzipping and builds one continuous strand.
- **Lagging strand:** On the other template, the enzyme must move away from the fork. It has to jump back to a new primer again and again, so the strand is made discontinuously as short pieces called **Okazaki fragments**.

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*Replication fork: leading and lagging strands*

### 4. Termination

Replication ends at a specific sequence called the termination site. The region between the origin site and the termination site is called a replication unit or **replicon**. **DNA polymerase I** removes the RNA primers and fills the gaps with DNA nucleotides. **DNA ligase** joins the Okazaki fragments of the lagging strand.

## 22. Explain the mechanism of transcription.

Transcription is the first step of gene expression, in which a complementary RNA is synthesised from DNA. Only one strand of DNA, the **template strand**, is transcribed. The other strand is the **coding strand**; it has the same sequence as the mRNA, except that T takes the place of U.

### 1. Initiation phase

No primer is needed. The enzyme **RNA polymerase** binds to the template DNA at a special nucleotide sequence called the **promoter**, which is not itself transcribed. In prokaryotes the promoters are TTGACA (–35 sequence) and TATAAT (–10 sequence). In eukaryotes they are TATAAA (TATA box, –25 sequence) and GGCCAATCT (CAAT box, –70 sequence).

RNA polymerase has five subunits: two α subunits (bind regulatory proteins), a β′ subunit (binds template DNA), a β subunit (binds RNA nucleotides) and a σ subunit (recognizes the promoter and initiates synthesis). After the enzyme attaches to the promoter, the σ subunit is removed. The other four subunits (core enzyme) unwind the DNA helix and form the **transcription bubble**.

### 2. Elongation phase

Inside the transcription bubble, RNA polymerase makes a chain of complementary ribonucleotides along the template strand in the 5′ to 3′ direction, usually starting with A or G. The nucleotides are added as ribonucleoside triphosphates (rNTP); two phosphates detach and release energy. The first 12 bases of the new mRNA make a temporary helix with the template DNA, which keeps the 3′ end of the mRNA at the catalytic site. The bubble moves down the DNA, the growing mRNA protrudes from it, and the transcribed DNA is re-wound behind it.

### 3. Termination phase

At the end of the gene there is a stop region: a series of GC base pairs followed by a series of AT pairs. When it is transcribed, hydrogen bonds form within the mRNA and make a loop called the **GC hairpin**, followed by a tail of four or more U ribonucleotides. The hairpin makes RNA polymerase pause. The pairing of the U's of mRNA with the A's of DNA is the weakest, so it cannot hold the RNA–DNA hybrid together. The RNA strand dissociates from the DNA and transcription stops.

## 23. Describe the mechanism of protein synthesis.

In translation (protein synthesis), the information present in an mRNA transcript is used to direct the sequence of amino acids of a polypeptide.

### Activation of amino acids

Specific tRNAs attach with specific amino acids and make aminoacyl-tRNAs (aa-tRNA). The enzyme **aminoacyl-tRNA synthetase** carries out this reaction; there are 20 kinds of it, one for each amino acid. The aa-tRNA delivers the amino acid to the ribosome.

### 1. Initiation phase

1. The aa-tRNA carrying the first amino acid binds to the smaller ribosomal subunit under the control of an initiation factor. The first amino acid is N-formyl-methionine in prokaryotes and methionine in eukaryotes.
2. Another initiation factor attaches the 5′ end of mRNA to the smaller subunit and positions the anticodon of the aa-tRNA over the first codon (AUG).
3. The larger ribosomal subunit is placed upon the smaller subunit. This forms the **initiation complex**.

The ribosome has three sites: the **P site** (peptidyl site, where peptide bonds form), the **A site** (aminoacyl site, where the next aa-tRNA binds) and the **E site** (exit site, where empty tRNAs leave). They are oriented 5′ to 3′ as E–P–A.

### 2. Elongation phase

1. The first aa-tRNA occupies the P site. The next aa-tRNA, with a matching anticodon, binds to the A site with the help of an elongation factor.
2. The enzyme **peptidyl transferase** removes methionine from the first tRNA at the P site and attaches it by a peptide bond to the amino acid of the second aa-tRNA at the A site.
3. The ribosome moves three nucleotides along the mRNA in the 5′–3′ direction. This movement is called **translocation**.
4. The tRNA carrying the chain of two amino acids shifts from the A site to the P site, a new codon is exposed at the A site, and the empty tRNA leaves from the E site.
5. These steps are repeated until a stop codon reaches the A site.

### 3. Termination phase

When a stop codon is exposed at the A site, no aa-tRNA comes there, because there is no anticodon for stop codons. The stop codon is recognized by a protein called the **release factor**. It releases the polypeptide from the tRNA at the P site. The tRNA is released and the ribosomal subunits separate from the mRNA.

## 24. Describe the difference between protein synthesis in prokaryotes and eukaryotes.

| Prokaryotes | Eukaryotes |
|---|---|
| Genes lack introns. | Most genes possess introns. |
| After transcription, mRNA is translated immediately; there is no post-transcriptional modification. | mRNA is modified before translation: introns are cut out, a 5′ cap and a 3′ poly-A tail are added. |
| Translation occurs in the cytoplasm. | Translation occurs in the cytoplasm as well as on the rough endoplasmic reticulum. |
| Ribosomes are smaller (70S). | Ribosomes are larger (80S). |
| The initial amino acid is modified, i.e., N-formyl methionine. | The initial amino acid is methionine; it is not modified. |

## 25. Describe the importance of the regulation of gene expression.

All the genes of an organism are not expressed at the same time, and all genes are not turned ON in all cells. Turning genes ON and OFF and the timing of their expression are controlled by regulator proteins; this is called the **regulation of gene expression**. Its importance:

- It enables cells to respond to environmental conditions and to prepare a protein only when it is needed.
- In multicellular organisms, it is the basis of cellular differentiation and morphogenesis. It creates different cell types: the genome is the same, but different genes are expressed in each type.
- Example: liver cells express the genes for the enzyme alcohol dehydrogenase, while neurons of the brain keep these genes turned OFF.

## 26. Describe the negative and positive control of gene expression.

At the transcriptional level, genes have special regulatory sequences to which specific regulatory proteins bind.

**Negative gene regulation:** Regulatory proteins called **repressors** bind to pieces of DNA called **operators**, present near the promoter. Here the repressors block RNA polymerase and stop or reduce transcription.

**Positive gene regulation:** Regulatory proteins called **activators** (or inducers) bind to the operators. They start transcription or increase its speed by helping RNA polymerase to bind to the promoter.

## 27. Explain the post-transcriptional modification of mRNA.

In prokaryotes, transcription occurs in the cytoplasm, so the new mRNA is translated directly. In eukaryotes, the new mRNA has to travel from the nucleus to the cytoplasm. For this journey it undergoes the following modifications:

1. **Addition of 5′ cap:** A methylated GTP is attached to the first nucleotide (A or G) at the 5′ end. This 5′ cap protects the mRNA from nucleases and phosphatases.
2. **Addition of 3′ poly-A tail:** A chain of about 250 A nucleotides is added at the 3′ end. It protects the mRNA from degradation by nucleases.
3. **Splicing:** The new mRNA is immature because it contains **introns** (regions that do not code for proteins) and **exons** (regions that are expressed). In splicing, the introns are removed and the remaining exons are joined. So the mature mRNA consists of coding segments only.

## 28. Write a note on point mutations and their sources.

A change in the content of the genetic message is called a mutation. Mutations involving a change in only one or a few nucleotides are called **point mutations**. Common examples:

1. **Substitutions:** exchange of a single nucleotide for another, e.g., A is substituted with G.
2. **Insertions:** addition of one or more extra nucleotides into the DNA, usually due to errors during replication.
3. **Deletions:** removal of one or more nucleotides from the DNA.

### Sources of point mutations

1. **Physical damage to DNA:** *Ionizing radiation* (X-rays, gamma rays) changes atoms into free radicals, which may break the phosphodiester bonds of DNA. *Ultraviolet radiation* is strongly absorbed by DNA; a double covalent bond forms between two adjacent pyrimidine nucleotides (C and T), which can block replication.
2. **Mutagen chemicals:** Some resemble nucleotides and are incorporated into DNA; some remove the amino group from adenine or cytosine; some add hydrocarbon groups to nucleotides. All cause mis-pairing. Example: mustard gas.
3. **Spontaneous point mutations:** Sometimes nucleotides spontaneously shift to alternative positions, and DNA polymerase pairs a different nucleotide with them. Sometimes sequences misalign when homologous chromosomes pair; a loop forms, the cell cuts it out, and several hundred nucleotides are deleted.

## 29. Describe the symptoms, causes and treatments Down syndrome and Klinefelter syndrome.

Occasionally, sister chromatids fail to separate properly in meiosis, so a gamete gains or loses a chromosome. This is called **non-disjunction**. It can happen in autosomes or in sex chromosomes.

In **Down syndrome**, the total chromosome number is 47 (2n+1). It was first described in 1866 by J. Langdon Down. About 1 in every 750 children has it.

| | Down syndrome |
|---|---|
| **Causes** | Non-disjunction in autosome pair 21 during the formation of the egg. The egg has 24 chromosomes; when a normal sperm fertilizes it, the child (male or female) has trisomy 21 with 47 chromosomes. |

| | Down syndrome |
|---|---|
| **Symptoms** | Mental impairment, stunted growth, increased skin on back and neck, flat facial features, small head, short neck and limbs, upward slanting eyes, small ears, poor muscles, protruding tongue and heart defects. Males are usually sterile; females have lower fertility. |
| **Treatment** | Proper management improves the quality of life: screening for common problems, medical treatment where indicated, a good family environment, special education and proper care. |

**Klinefelter syndrome** is the XXY condition (47 chromosomes; 2n+1 or 44+XXY). It was first described by Harry Klinefelter in 1942.

| | Klinefelter syndrome |
|---|---|
| **Causes** | Non-disjunction of the X chromosomes in the mother gives an XX egg; when a normal Y sperm fertilizes it, the child is XXY. Non-disjunction in the father can give an XY sperm, which with a normal X egg also gives XXY. |
| **Symptoms** | Sterile males with many female body characteristics (less body hair, breast growth), small testicles, abnormal body proportions (long legs, short trunk, shoulders equal to hip size); the voice may not be as deep. |
| **Treatment** | Testosterone (to compensate its deficiency at puberty); surgical removal of breast; behavioural therapy for language disorders, difficulties at school and socialization. |

## 30. Describe the symptoms, causes and treatments Turner syndrome.

**Turner syndrome** is the XO condition in a female (45 chromosomes; 2n–1 or 44+XO), i.e., monosomy in the sex chromosomes. It was first described by Dr. Henry Turner in 1938. It occurs roughly once in every 5000 female births.

| | Turner syndrome |
|---|---|
| **Causes** | Non-disjunction gives an egg (or sperm) with no sex chromosome (O). If an O egg fuses with a normal X sperm, or an O sperm with a normal X egg, an XO zygote is formed. (An OY zygote cannot develop.) |
| **Symptoms** | The female is sterile, with short stature, webbed neck, immature sex organs, low mental abilities, and small chin and jaw. Associated problems: liver and kidney issues, obesity, diabetes and hypertension. |
| **Treatment** | Growth hormone treatment started early in childhood can give adult height; estrogen replacement therapy for the development of secondary sexual characteristics. |

## 31. Describe the symptoms, causes and treatments of Sickle cell anaemia and phenylketonuria.

**Sickle cell anaemia** is due to a recessive gene for an abnormal β chain of haemoglobin, present on chromosome 11.

| | Sickle cell anaemia |
|---|---|
| **Causes** | A substitution-type point mutation: the DNA nucleotide T is substituted by A. The codon of glutamic acid (CTC) changes to CAC, which is for valine. A child who receives both mutant alleles makes abnormal β chains. |
| **Symptoms** | The abnormal haemoglobin carries less oxygen and the RBCs become sickle-shaped. Pain, fatigue, shortness of breath, jaundice and an increased risk of infections. |
| **Treatment** | Pain management with NSAIDs and opioids; blood transfusions to increase healthy RBCs; bone marrow transplant for a long-term cure; vaccinations and antibiotics to prevent complications. |

**Phenylketonuria (PKU)** is a genetic metabolic disorder. A normal gene on chromosome 12 produces the liver enzyme phenylalanine hydroxylase (PAH), which converts the amino acid phenylalanine into tyrosine.

| | Phenylketonuria |
|---|---|
| **Causes** | A point mutation (substitution or deletion of a nucleotide) makes the allele recessive. A child with both recessive alleles does not make PAH properly, so phenylalanine builds up to toxic levels in blood and tissues. |
| **Symptoms** | Intellectual disability, seizures, behavioural problems, mental disorder, microcephaly (smaller head size), musty odour in breath, skin and urine; brain damage. |
| **Treatment** | Avoid foods high in phenylalanine (meat, dairy products, eggs, some grains); regular monitoring of blood phenylalanine; medicines containing synthetic PAH enzyme; nutritional supplements. |

## INQUISITIVE QUESTIONS

### 7. DNA is tightly wrapped around histone proteins. How can replication enzymes read the code if it is buried inside a coil?

The DNA is not read while it is coiled. Before replication, the chromosome is in its uncoiled chromatin form (interphase). At the replication origin, the DNA is loosened from the histones and helicases break the hydrogen bonds between the two strands. The bases of each single strand are then exposed, and single-strand binding proteins keep them open for DNA polymerase. After the fork passes, the DNA wraps around histones again (new histones are also made in interphase).

*The loosening of DNA from histones during replication is not described in the textbook; this part is based on general knowledge.*

**8. Our genetic code uses three-letter "words" to create 20 amino acids. Could a complex human exist if the code used only two-letter words?**

No. With four kinds of nucleotides, two-letter codons give only 4 × 4 = 16 combinations. These cannot specify 20 amino acids (and stop signals are also needed). With at most 16 amino acids, the thousands of different proteins of the human body could not be made. Three-letter codons give 4 × 4 × 4 = 64 combinations, which are more than enough.

**9. DNA Polymerase III is a highly efficient proof-reader that fixes its own mistakes. What happens to a cell if the gene for the "proof-reader" enzyme itself gets a mutation?**

The stability of DNA is lost. The faulty enzyme cannot correct wrongly paired nucleotides, so replication errors remain in the new DNA. With every division, mutations collect in many genes of the cell and its daughter cells. Most mutations are harmful, so the cell may make non-functional proteins and die, or its gene regulation may be disturbed, which can lead to diseases such as cancer.

**10. Why the length of transcribed mRNA molecule (in eukaryotes) shortens as it enters the cytoplasm for translation?**

The newly transcribed mRNA is immature. It contains introns (non-coding regions) as well as exons (coding regions). Before the mRNA leaves the nucleus, the introns are removed and the exons are joined; this is called splicing. So the mature mRNA that enters the cytoplasm is shorter than the transcribed mRNA.

**11. How many types of tRNA molecules are necessary for a living cell, if the genetic code is a triplet code?**

A triplet code has 64 codons. Three of them (UGA, UAG, UAA) are stop codons and have no anticodon, so no tRNA is needed for them. The remaining 61 codons specify amino acids. So, with one tRNA for each of these codons, **61 types of tRNA** are needed. In any case, at least 20 types are necessary, one for each of the 20 amino acids.

*In actual cells fewer than 61 tRNAs are enough, because one tRNA can read more than one codon of the same amino acid (wobble). This is beyond the textbook.*

**12. After isotopic labelling of the starting DNA with $^{15}N$, Meselson and Stahl continue their experiment to at least two rounds of replication instead of stopping the experiment after only one round of replication. Justify their decision to continue to at least two replications.**

After one round, all the DNA made a single band of intermediate density. This result rejects only the conservative model (which predicts one heavy and one light band). But the semi-conservative model and the dispersive model both predict intermediate DNA after the first round, so one round cannot separate them.

| Model | After 1st replication | After 2nd replication |
|---|---|---|
| Conservative | One heavy + one light band | One heavy + one light band |

| Model | After 1st replication | After 2nd replication |
|---|---|---|
| Semi-conservative | One intermediate band | One intermediate + one light band |
| Dispersive | One intermediate band | One band only (between intermediate and light) |

After the second round they found two bands, one intermediate and one light. This matches only the semi-conservative model. So the second round was necessary to prove it.

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